Farruh **Kushnazarov**

Senior Data Scientist

China/Shanghai

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My favorite subjects are mathematics and communication network system. Because of this, related to the communication network system. As a researcher, I not only considered in the theoretical part but worked with several companies for applications as well. Moreover, each result was provided for implementations on data-centers, smart-city, and other projects. In Russia, especially Saint-Petersburg researcher, my papers were frequently cited and discussed in dozens of conferences by other researchers. Every year, I publish scientific articles on artificial intelligence and communication network fields in international conferences, including the global academic platforms such as IEEE, ACM, and others.

Experience

Big Data & Data Scientist Manager

Shanghai, China

Ping An Hao Xue, under Ping An (平安) China

September 2016 –> Till now

- Languages — Python, R, RMarkdown, SQL, Spark and Hive
- NLP: have done projects like text classification, sentiment analysis, and text summarization.
- Implemented and retrained Mozilla DeepSpeech library for Automatic Speech Recognition
- Created and tested the Speech Accent Classification System for native and non-native speakers, with a rate of 99% in metric recall
- Conducted classification analyses of the customer life cycle stage to increase overall turnover
- With algorithmic, optimized ongoing ML/DL models and checked the performance of implemented models
- Established the Machine Learning model for refund customers and updated the refund model. Decreased the overall refund rate to 27%

Data Scientist (Consultant)

Online

PeriPlus AG (Part-Time)

November 2018 –> July 2019

- Languages — Python, R, RMarkdown and SQL
- Implemented AI/ML/DL projects
- From scratch established the data scientist team in three months
- Constructed a Container Number Recognition System using object detection and character recognition models
- Built the Face Recognition System for customers

Automatic Driving Prospective Technology Engineer

Shanghai, China

Haima automatic Investment Group Co. Ltd, R&D Center

April 2016 –> September 2016

- Languages — C/C++, Python and Bash
- Connected lidar and radar to decrease noises in raw data
- Designed a model car to check the capability of algorithms on ultrasonic sensors
- Optimized CANBus protocol to increase the efficiency of data transfer
- Optimized joint work and logic compatibility of equipments

Senior Programmer, Data Analyst

St Petersburg, Russia

Domus Sapiens

January 2011 –> May 2015

- Languages — C/C++, Assembler and Bash
- Lead and managed more than 20 projects on intelligent systems. Some of the most famous and complicated cases are the Moscow Smart Lighting Projects
- Moscow Smart Lighting Projects were to do with five high buildings with various styles and functions, including office buildings and hotels. The requirements were not even similar from floor to floor, which made the whole projects equivalent to establish a small smart city system. We successfully finished the projects and because of this, Domus Sapiens got several rewards
- Implemented and applied the patent model to improve the quality of data transfer in wireless networks

Software Engineer

Tashkent, Uzbekistan

ProDVD

October 2010 –> March 2012

- Languages — C/C++
- Developed the Card King for mobile phones, compatible for both Android and iOS systems, making it convenient for business people to organize and store business cards
- Developed the audio encoding/decoding part for the Blu Ray player
- Invented the word recognition software, both iOS and Android compatible

Assistant of Professor *Tashkent, Uzbekistan*

Tashkent Railway Engineering Institute *September 2010 –> July 2012*

- To conduct laboratory and practical classes for below subjects:
- Programming in C/C++ languages;
- Network & communication systems on the railway;
- Information systems on railway transport
- To assist the professor in research work, create application parts, and simulation parts of analytical models

Education

Emperor Alexander I St.Petersburg State Transport University *St Petersburg, Russia*

Mathematical modeling, numerical methods and computer programs *2012 –> 2016*

- Doctor of Philosophy (Ph.D.)
- Research Field: Develop data transmission methods for evaluating the real speed of data link layer protocols, which aimed at improving hardware and software components

Emperor Alexander I St.Petersburg State Transport University *St Petersburg, Russia*

Information systems and technologies *2008 –> 2010*

- Degree: Master of Science
- Research Field: Develop and optimize computer network systems

Emperor Alexander I St.Petersburg State Transport University *St Petersburg, Russia*

Information systems and technologies *2004 –> 2008*

- Degree: Bachelor of Computer Science
- Research Field: Develop and optimize computer network systems

Data Science Skills

Communication

Presentations (technical and non-technical), dashboard design, data analysis reports, scientific publications and data visualization

Programming/Markup Languages

Python, R, SQL/NoSQL, Hive, Spark, C/C++, Matlab, BASH, CSS, HTML, LaTeX, Markdown and RMarkdown

Software Development

Docker, git, version control, automated testing and continuous integration

Numerical Methods

optimization (stochastic, genetic, multi-start) and numerical solution of differential equations

Statistics

Machine learning, data analysis, generalized linear regression, cluster analysis, factor analysis, principal components analysis (PCA), cross validation, generalized additive models, data analytics and Monte Carlo simulation

Environments

Linux, Jupyter-Lab, PyTorch, RStudio, ETL, SQL Server Management Studio, Azure Data Studio, VIM and Visual Studio

Publications

1. Kushnazarov, F (2019). Data Stream Controlling in Communication Channels with Noise. *2019 IEEE 4th International Conference on Cloud Computing and Big Data Analysis (ICCCBDA)*, 534–538.
2. Kushnazarov, F et al. (2019). *Consumer Life Cycle and Profiling: A Data Mining Perspective*. IntechOpen. `https://www.intechopen.com/books/consumer-behavior-and-marketing/consumer-life-cycle-and-profiling-a-data-mining-perspective`.
3. Kushnazarov, F (Apr. 2018). Throughput of communication protocols for distributed systems transferring a group of frames under noise. *2018 IEEE 3rd International Conference on Cloud Computing and Big Data Analysis (ICCCBDA)*, 424–429.
4. Кушназаров, ФИ and ВГ Бабина (2018). НАПРАВЛЕНИЯ РАЗВИТИЯ КОРПОРАТИВНОЙ ИНФОРМАЦИОННОЙ СИСТЕМЫ КОМПАНИИ ”УЗБЕКИСТОН ТЕМИР ЙУЛЛАРИ”. *Интеллектуальные технологии на транспорте* (3 (15)).

5. Kushnazarov, F (2017). Analysis of perfomance of algorithms for scoring system in organization of customer profiles. *2017 IEEE 2nd International Conference on Big Data Analysis (ICBDA)*, 281–285.
6. Yakovlev, V and F Kushnazarov (Oct. 2017). *Organization of work with cloud services*. 1st ed. 190031, Saint Petersburg, 9 Moskovsky pr.: FGBOU VO PGUPS, p. 43.
7. Kushnazarov, F (2016). "Development and research of models of performance evaluation of communication protocols for channel with noises". PhD thesis. St. Petersburg: Emperor Alexander I St. Petersburg State Transport University, p. 159.
8. Kushnazarov, F (2016). Analysis of the data transmission in communication channels with noise. *Prace Naukowe Politechniki Warszawskiej. Transport*.
9. Kushnazarov, F (Feb. 2015). Controlling data flows in the noisy channels. *Intellectual Technologies on Transport.* **1**(1).
10. Kushnazarov, F and V Yakovlev (Oct. 2015). "The Protocols Performance Evaluation of the Data Link Layer in the ISO/OSI Model". Patent RU 2015619739 (RU). `https://patentinform.ru/programs/reg-2015619739.html`.
11. Yakovlev, V and F Kushnazarov (Dec. 2015). Evaluation of the effect of interferences on link-layer protocol performance. *Proceeding of Petersburg Transport University* **1**(42), 133–138.
12. Кушназаров, ФИ, ВВ Яковлев, and ОА Турдиев (2015). Сравнение производительности протоколов доступа к облачным ресурсам. *Известия Петербургского университета путей сообщения* (4 (45)).
13. Кушназаров, ФИ, ОА Турдиев, and ФИ Кушназаров (2015). ЭФФЕКТИВНОСТЬ ЦИКЛИЧЕСКИХ КОДОВ ПРИ ПЕРЕДАЧЕ ДАННЫХ В КАНАЛАХ СВЯЗИ С ПОМЕХАМИ. *Сборник трудов VII научно-практической конференции молодых ученых* (73).

Projects

NLP: Speech Accent Detection

Project Link

Everyone who speaks a language, speaks it with an accent. This project defines accents for the english language speakers

2020

- Role: Author and Maintainer
- Results: Accuracy=.90, Recall=.91 and Precision=.93
- Increase accuracy of ASR (Automatic Speech Recognition)

NLP: Text classification

Private

Classified to 10 different topics

2019

- Data sources: Title, Description, Text
- Method: LDA
- Result: 10 topics. Accuracy=.87

Moscow Smart Lighting Projects

Project Link

Moscow Smart Lighting Projects Project, which is made by 5 different office building groups with different requirements to smart system and linking. Companies main job is to make computer programming of center control and night light adjustment. More then 800 square meters, with 15000 signals from different controllers

2015

- Maintainer

L-Cube

Company WebPage

Implemented at the company L-Cube LLC in order to optimize the performance of the departmental network

2015

- Author and Maintainer

EMC-DELL

Implementation Certificate

Certificates from EMC2 Data-Center about Value of Research and Possibility of Implementation the Model

2016

- Author and Maintainer

IntellektTrans

Conference WebPage

Intelligent rail system transportation summit. As one of the members of the Organizing Committee of the intelligent rail system transportation summit, the organization participated in the summit and attended the meeting

2009

- Volunteer

Accomplishments

2020	Data Analysis with Python	*IBM and Coursera*
2020	Data Science Methodology	*IBM and Coursera*
2020	Tools for Data Science	*IBM and Coursera*
2020	What's Data Scientists	*IBM and Coursera*
2017	Big Data 101	*Big Data University*
2016	Data Mining	*Intuit National Open University*
2013	Certificate of knowledge of technology AMX I - II level	*Smart City and IoT*
2013	Certificate of knowledge of ABB technology I-III	*Smart City and IoT*
2013	Certificate knowledge of technology Crestron I - II level	*Smart City and IoT*
2012	Data Science - effective use of data	*DELL-EMC*
2012	ISM - administration of information and data storage	*DELL-EMC*
2012	The C Programming Language	*Intuit National Open University*
2007	Fundamentals of IBM z Series	*IBM*